

1. Title of Proposed Idea/Innovation

FIRENET SENTINEL PODS

(MSME-led Smart Forest Fire Detection and First Response System)

2. Problem Statement

Forest fires in India destroy biodiversity, release massive CO₂, and affect lakhs of hectares annually. Existing systems like satellites and drones are expensive, delayed, and ineffective for forest-edge regions. There's no affordable, on-ground real-time mechanism for early detection and suppression.

FireNet Sentinel Pods fill this gap. These solar-powered pods detect fire indicators and send instant SMS/email alerts with GPS location. Their key innovation is a built-in electromechanical first-response system that automatically releases sand, water mist, or fire-retardant gel when critical thresholds are crossed. Local MSMEs will build and maintain these systems, creating rural employment and climate resilience.

3. Description of the Project

The pod is a solar-powered, waterproof unit with heat, gas, and smoke sensors. When triggered, it sends alerts to authorities, logs data to a cloud dashboard, and activates a local suppression system — sand hopper, water mist, or gel cartridge — via relay or servo motor.

Powered by an ESP32 microcontroller and GSM/LoRa network, the system works in offline regions. It operates autonomously with minimal maintenance. Local MSMEs assemble the units and receive training in deployment, repair, and monitoring. The project integrates AI to predict fire spread and identify hotspots. Target areas include forest-edge tribal belts, plantations, parks, and reserves.

4. Technical Details of the Project

Use Cases:

- Forest-edge tribal villages
- Bamboo or eucalyptus plantations

- Wildlife zones and eco-tourism parks

Technologies Used:

- ESP32 MCU, GSM/LoRa, GPS
- Smoke, gas, heat sensors
- Solar power, AI dashboard, relay/servo/solenoid control

Response Components:

- Water tank + DC pump
- Sand flap + servo
- Gel canister + solenoid

Sustainability Impact:

- CO₂ mitigation through fire control
 - New rural MSME business streams
 - Ecosystem and wildlife protection
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5. Uniqueness of the Innovation

- India's first low-cost pod with automated local fire response
 - Works offline without internet
 - MSME-friendly production and servicing
 - Easily replicable across fire-prone zones
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6. Objectives & Application

- Build and test 50 units
 - Deploy in 3 states
 - Train 5 MSMEs
 - Target <2 min response time
 - Use in Nilgiris, Wayanad, MP, Odisha forests, plantations, parks
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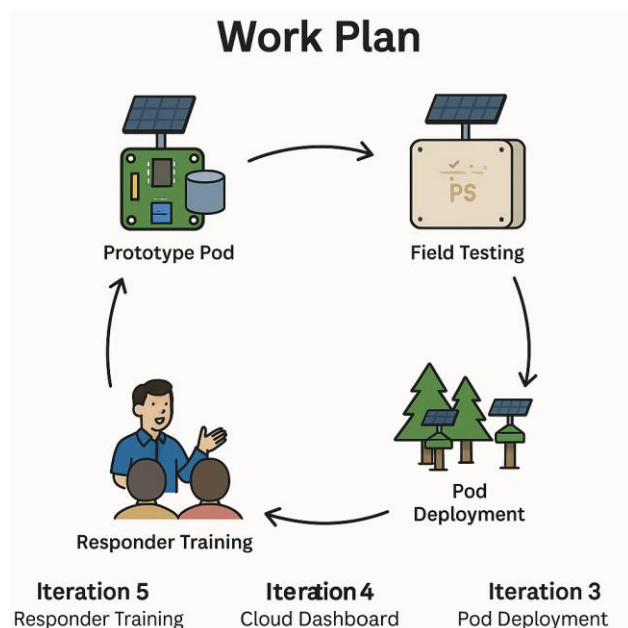
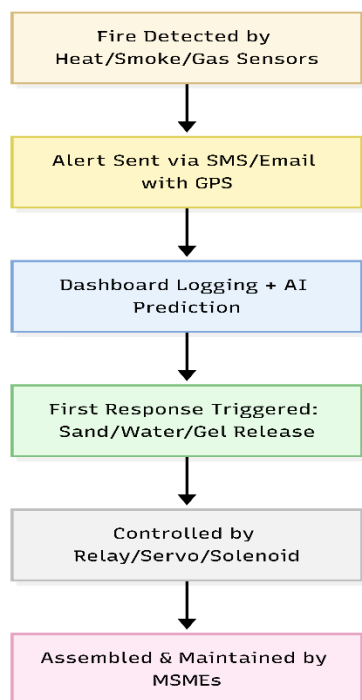
7. Market Potential

India's 1L forest-edge villages $\times$ 10 pods = 1M unit demand. ₹500–1,000 Cr market potential in India. Global demand in SE Asia, Africa. Buyers: forest departments, CSR programs, NGOs.

Sources: FSI 2023, MoEFCC, Global Forest Watch

8. Prototype (If Available)

- ESP32 prototype with heat/smoke sensors
- Working alert system + fire-response mockup
- Solar-powered shell under build



9. Expected Outcomes

- 70–90% early containment in pilots
- 2 min average detection-to-response time
- 100+ MSME jobs created
- First MSME-made wildfire first-response system in India